Impact Of Transforming Changes In **Vehicle Architecture**

Smart mobility will change the way we commute, and with this the hardware and software associated with the vehicle will also revolutionise. Here is an expert opinion on how the automotive trends will affect the vehicle architecture

Vinayaka Nagaraja

Automotive technologies are consistently progressing. Technologies for connected mobility and its infrastructure are already hitting the market. With these advancements, there is going to be an ever-growing need for computation power inside the vehicle and the cost will grow exponentially. To tackle this challenge, there is a need for newer vehicle architecture that can handle data and functions with minimum control units.

In this article, we will look at some trends in the automotive industry that are emerging with recent technologies, and how improved vehicle architecture can cater to all the recent requirements.

Trends revolutionising the architecture

We are all aware of the recent advancements in automotive industry, which are only going to improve further. So, technologies inside a car, its architecture, and the electronic control units (ECUs) have all to be aligned for this radical shift in the way we commute.

Besides, we know that alternative drives are taking over, and we are going beyond the traditional combustion engines. We are looking at alternate ways that could make our mobility sustainable and also leave less carbon footprints in the environment. This is influencing the way the technologies in vehicles are being imagined today.

Cloud services are an indispensable part of connected vehicles to pull-in necessary software and updates directly without interference. With connectivity, a vehicle and its architecture would be a part of the mobility ecosystem. The automotive technology has so far lagged behind and has to catch up and be prepared for the IoT revolution happening in the ecosystem.

We are in the era of tablets and smart-

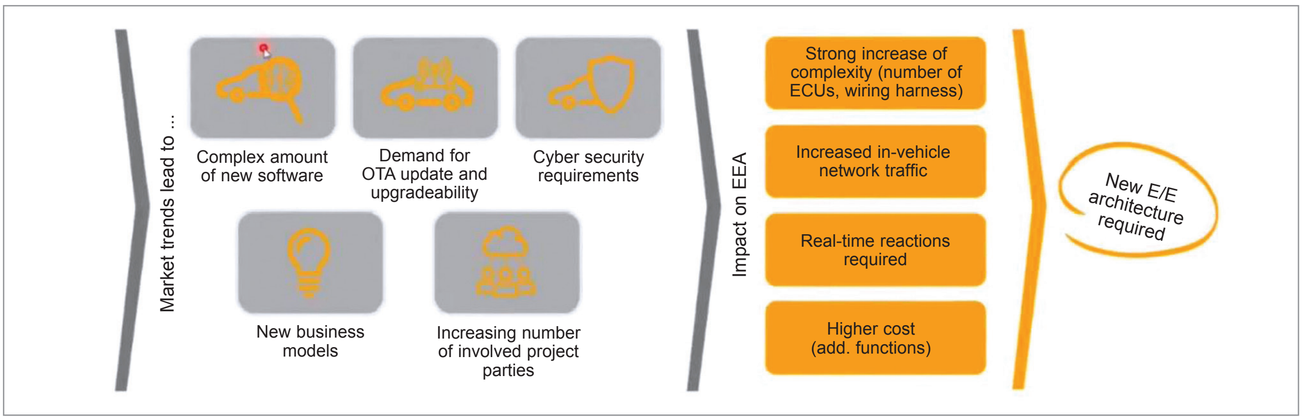

Fig. 1: Upcoming market trends lead to increase in software complexity and the need for new E/E architecture